GRT Seminar Spring 2026 – Potent Categorical Representations

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January 31, 2026

Abstract

This semester, the GRT Seminar will focus on potent categorical representations, following the 2026 ICM address of Nadler–Ben-Zvi (based on joint work with Stefanich). There are many open questions!

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1 (1/22) David Nadler – Introduction

We're going to start with the first page or so of the paper. The rough idea here is to instantiate some of the ideas in Constantin Teleman's ICM address.

1.1 Categorical representations

Our motivation for studying categorical representations comes from the following analogy.

If Γ is a finite group, then we look at the group ring $\mathbb{C}[\Gamma]$ We may think of this as a ring of distributions, with multiplication given by convolution. Studying representations of Γ corresponds to studying modules over $\mathbb{C}[\Gamma]$.

In general, if $\mathcal{C}$ is a symmetric monoidal (∞-)category, e.g. $(\mathsf{Vect}_{\mathbb{C}}, \otimes)$, we may consider an algebra object $\mathsf{A} \in \mathcal{C}$. We can study the category of modules $\mathsf{Mod}_{\mathsf{A}}(\mathcal{C})$.

We can specialize this very general framework to give a notion of categorical representations. Let G be a connected reductive group over $\mathbb{C}$.

Definition 1.1. A *categorical* G*-representation* is a (presentable) stable $\mathbb{C}$-linear ∞-category which is a module over $\mathsf{D}(\mathsf{G})$, the derived ∞-category of D-modules on G. We'll write $\mathsf{G}\text{-}\mathsf{Cat}$ for the ∞-category of categorical G-representations.

Remark 1.2. Morally, the reason we use D-modules here is because they quantize the cotangent bundle. The really important object is $\mathsf{T}^*\mathsf{G}$ (with two moment maps $\mathsf{T}^*\mathsf{G} \rightrightarrows \mathfrak{g}^*$ corresponding to the left and right actions of G). Everything here is secretly symplectic!

Example 1.3. Let $\mathsf{G} \curvearrowright \mathsf{X}$. Then we get an action $\mathsf{D}(\mathsf{G}) \curvearrowright \mathsf{D}(\mathsf{X})$. This really corresponds to the moment map $\mathsf{T}^*\mathsf{X} \to \mathfrak{g}^*$.

1.2 Categorical representations of $\mathbb{G}_m$

Let's see what happens when $G = \mathbb{G}_m$.

Theorem 1.4. *There is a monoidal equivalence* $D(\mathbb{G}_m) \simeq QC(\mathbb{A}^1/\mathbb{Z})$.

Indeed, recall that a D-module on $\mathbb{G}_m$ comes with two pieces of data:

- An action of the Euler vector field $t\partial_t$, corresponding to the coordinate on $\mathbb{A}^1$.
- An action of t and t^{-1}, corresponding to translation-equivariance.

This correspondence sends the δ distribution at 1 to the structure sheaf.

Corollary 1.5. *There is an equivalence* $\mathbb{G}_m\text{-}\mathsf{Cat} \simeq QC(\mathbb{A}^1/\mathbb{Z})\text{-}\mathsf{Mod}$.

Reasonable objects within $QC(\mathbb{A}^1/\mathbb{Z})\text{-}\mathsf{Mod}$ have "singular support" in (the 2-shifted version of) $T^*(\mathbb{A}^1/\mathbb{Z})$. This translates into a notion of "position" and "momentum" for such objects.

Example 1.6. If we consider a point $\lambda \in \mathbb{A}^1/\mathbb{Z}$, the module category $QC(\lambda)$ corresponds to

$$D_\lambda(\mathbb{G}_m) = \left\langle \mathscr{O}_{\mathbb{G}_m}, d - \frac{\lambda}{z} dz \right\rangle \subset D(\mathbb{G}_m).$$

Roughly: the position of a $QC(\mathbb{A}^1/\mathbb{Z})$-module corresponds to the monodromy. This relates to Langlands parameters.

Momentum should correspond to "equivariant parameters." To capture this, we must work microlocally. This is the motivation for considering potent categorical representations.

1.3 Another motivation

Sufficiently dualizable 2-categories give rise to 3d TQFTs. One can obtain some 3d TQFTs by taking S^1-compactifications of 4d TQFTs, i.e. setting $Z_3(\mathrm{pt}) = Z_4(S^1)$. If Z_4 is pure gauge theory, Z_3 corresponds to considering gauge theory with classical fields G/G. Remember that G/G can be thought of as the free loop space $\mathscr{L}BG$. The "equivariant parameters" here live in the numerator G, while the "monodromies" / "G-representations" come from the denominator G.

1.4 Defining potent categorical representations

We can write $G\text{-}\mathsf{Cat} = 2\mathsf{IndCoh}(G_{dR})$. Thus a possible extension is:

Definition 1.7. Let $G\text{-}\mathsf{Cat}^{\mathrm{pot}} = 2\mathsf{IndCoh}(G/G)^{\mathrm{Tate}}$.

This contains $G\text{-}\mathsf{Cat}$ as the full 2-subcategory set-theoretically supported at $1 \in G$. There are other subtleties (e.g. shearing, genuine equivariance) that should appear. We end up with a theorem:

Theorem 1.8. *There is an equivalence* $\mathbb{G}_m\text{-}\mathsf{Cat}^{\mathrm{pot}} \simeq 2\mathsf{IndCoh}(\mathbb{A}^1/\mathbb{Z})$.

We may also define a Lie algebra version:

Definition 1.9. Let $G\text{-}\mathsf{Cat}^{\mathrm{pot,gr}} = 2\mathsf{IndCoh}(\mathfrak{g}/G)^{\mathrm{Tate}}$.

Our goal this semester is to understand these definitions and theorems.

2 (1/29) Samuel Mayonnaise – Loop Spaces and D-Modules

Our goal for today is to prove the following.

Theorem 2.1. *Let* X *be a smooth scheme. Then there is an equivalences*

$$D(X) \otimes k[u, u^{-1}] \simeq \mathsf{IndCoh}(\mathcal{L}X)^{\mathrm{Tate}}.$$

The "Tate construction" here captures an S^1-localization. To understand the above statement, we will need to recall some relevant definitions.

2.1 Loop stacks

Let X be a derived stack, i.e. a (covariant) functor from derived commutative rings to spaces satisfying a sheaf condition.

Remark 2.2. There are many models for "derived commutative rings" that we can use. In characteristic zero we may model derived commutative rings by connective commutative differential graded algebras. In general we can work with simplicial commutative rings.

The *loop stack* of X is $\mathcal{L}\mathsf{X} = \underline{\mathrm{Hom}}(\mathsf{S}^1, \mathsf{X})$. Here S^1 is viewed as a derived stack by sheafifying the constant sheaf with value S^1. More generally, sheafifying constant sheaves gives a definition of the *Betti stack* of any topological space.

Remark 2.3. The simplicial perspective can be useful in understanding this. We may think of the Betti stack of S^1 as a coequalizer of $\mathrm{pt} \rightrightarrows \mathrm{pt}$. This corresponds to building up S^1 as a simplicial set with nondegenerate 0- and 1-simplices.

Example 2.4. If X is a Betti stack, then $\mathcal{L}\mathsf{X}$ is the Betti stack corresponding to the (topological) loop space of X.

Let's think of S^1 as a (homotopy) pushout:

$$\begin{tikzcd} \mathsf{S}^0 \arrow[r] \arrow[d] & \mathrm{pt} \arrow[d] \\ \mathrm{pt} \arrow[r] & \mathsf{S}^1. \end{tikzcd}$$

This allows us to write

$$\mathcal{L}\mathsf{X} = \underline{\mathrm{Hom}}(\mathrm{pt} \sqcup_{\mathsf{S}^1} \mathrm{pt}, \mathsf{X}) = \underline{\mathrm{Hom}}(\mathrm{pt}, \mathsf{X}) \times_{\underline{\mathrm{Hom}}(\mathsf{S}^0, \mathsf{X})} \underline{\mathrm{Hom}}(\mathrm{pt}, \mathsf{X}) = \mathsf{X} \times_{\mathsf{X} \times \mathsf{X}} \mathsf{X}.$$

Thus we may interpret $\mathcal{L}\mathsf{X}$ as the derived self-intersection of the diagonal of X. This suggests that D-modules may potentially appear, since the normal bundle of the diagonal is the tangent bundle of X.

Assuming X has affine diagonal, we may write

$$\mathcal{L}\mathsf{X} = \underline{\mathrm{Spec}}_{\mathscr{O}_\mathsf{X}}(\mathscr{O}_\mathsf{X} \otimes_{\mathscr{O}_\mathsf{X} \otimes \mathscr{O}_\mathsf{X}} \mathscr{O}_\mathsf{X})$$

where all operations here are implicitly derived.

2.2 Hochschild homology

Suppose $\mathsf{X} = \operatorname{Spec} \mathsf{R}$ is affine. Then

$$\mathscr{O}(\mathcal{L}\mathsf{X}) = \mathsf{R} \otimes_{\mathsf{R} \otimes \mathsf{R}} \mathsf{R} = \mathsf{HC}_*(\mathsf{R})$$

the *Hochschild chains* of R. The *Hochschild homology* $\mathsf{HH}^*(\mathsf{R})$ is the homology of this complex. In general, Hochschild chains are a "derived version of the abelianization $\mathsf{R}/[\mathsf{R}, \mathsf{R}]$"

Remark 2.5. When X is a topological space / Betti stack, we have $\mathscr{O}(\mathsf{X}) = \mathsf{C}^*(\mathsf{X})$. The above discussion roughly translates into something like $\mathsf{H}^*(\mathcal{L}\mathsf{X}) = \mathsf{HH}_*(\mathsf{C}^*(\mathsf{X}))$, at least when X is simply connected (so the diagonal is affine).

Theorem 2.6 (Hochschild–Kostant–Rosenberg)**.** *Let R be a commutative algebra with $\mathsf{X} = \operatorname{Spec} \mathsf{R}$ smooth. Then there is an equivalence of dg-algebras $\mathsf{HC}_*(\mathsf{R}) \simeq \Omega^*_\mathsf{R} = \mathrm{Sym}^* \Omega^1_\mathsf{R}[1]$, the complex of differentials on R. Geometrically, this translates into $\mathscr{O}(\mathcal{L}\mathsf{X}) \simeq \Omega^*_\mathsf{X}$.*

Definition 2.7. Let $\mathbb{T}_\mathsf{X}[-1]$ be the odd tangent bundle of X, i.e. $\underline{\mathrm{Spec}}_{\mathscr{O}_\mathsf{X}} \mathrm{Sym}^* \Omega^1_\mathsf{X}[1]$.

HKR can be interpreted as an equivalence $\mathscr{L}\mathsf{X} \simeq \mathbb{T}_\mathsf{X}[-1]$ for any smooth scheme X. David pointed out that this equivalence should be thought of as an exponential map $\mathbb{T}_\mathsf{X}[-1] \to \mathscr{L}\mathsf{X}$. For stacks, the above statement of HKR fails.

Example 2.8. For $\mathsf{X} = \mathsf{BG}$, the loop space $\mathcal{L}\mathsf{G}$ is the adjoint quotient G/G. However, $\mathbb{T}_\mathsf{X}[-1]$ is $\mathfrak{g}/\mathsf{G}$, which is quite different.